Learning parity from selected examples under a globally separated neural kernel

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Abstract

We construct selected training sets on which a globally margin-conditioned neural Gaussian prior learns sparse parity using $d^{O(\sqrt{k})}$ examples. For any fixed error tolerance, every minimum-energy compatible field has that error or less, and all such fields have the same signs. The construction uses alternating interpolation on central Hamming layers and a general sparsification lemma for convex margin minimizers. It also produces actual global minimizers with very large amplitudes at some correctly classified inputs. These are deliberately selected examples, not independent uniform samples. The result does not settle the proposed $\exp\{\Omega(k\log(ed/k))\}$ lower bound for uniform random training; it shows that typical sampling is essential to such a bound. It suffices that $d \geq (C_L k)^{C_L k}$.

1 Model and result

Let $\mathcal{X}_d = \{-1,1\}^d$ have uniform probability measure. Fix $L \geq 2$, define $\Psi(t) = 2\arcsin(t)/\pi$, and put

$$F = \Psi^{\circ(L-1)}, \qquad K_d(x,y) = F\left(\frac{1+x\cdot y}{d+1}\right).$$

This covariance is strictly positive definite and has diagonal one. For a real field z on the cube, its energy is

$$\|z\|_K^2 = z^{\mathsf{T}} K_d^{-1} z.$$

The factor $1/2$ in Gaussian rate functions does not affect the minimizers. The unit margin entails no loss of generality: multiplying fields by a fixed margin $a > 0$ multiplies all energies by a^2 and preserves signs.

The target is $f(x) = \prod_{i=1}^k x_i$. Given a finite set of training inputs T, consider

$$\min_z \|z\|_K^2 \quad \text{subject to} \quad |z(x)| \geq 1 \ (x \in \mathcal{X}_d), \qquad f(x)z(x) \geq 1 \ (x \in T). \tag{1}$$

This is a global separation constraint at *every* input. A minimizer's error is $\mathbb{P}_x\{f(x)z(x) < 0\}$.

Theorem 1 (Selected examples)**.** *Fix $L \geq 2$ and $\delta \in (0, 1/2)$. For every sufficiently large odd k and every $d \geq (C_L k)^{C_L k}$, there is a parity-labeled training set $T \subseteq \mathcal{X}_d$ with*

$$|T| \leq d^{2\sqrt{k/\delta}+9}$$

for which all minimizers of (1) have the same sign function and error at most δ. At some correctly classified inputs, every such minimizer has amplitude at least

$$\exp\{c_\delta \sqrt{k}\log k\}.$$

The positive constant c_δ is independent of k, d; C_L depends only on L, and the threshold on k may depend on δ.

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The proof first constructs a convex margin minimizer from all inputs in a central Hamming window, proves that it already has the required global separation, and then reduces its training constraints by sampling its contact measure. This last sampling law is part of the construction.

The explicit dimension condition permits growing k up to a constant multiple of $\log d/\log\log d$. The training set is chosen using a contact measure of a convex margin problem. It is not distributed as independent uniform examples. Nevertheless, for fixed k the displayed number of examples is a vanishing fraction of the 2^d inputs. For growing k, its exponent in d is sublinear in k.

Corollary 2 (Actual label probabilities)**.** *For the selected set in Theorem 1, let Z_t be Gaussian with covariance tK_d, conditioned on $|Z_t(x)|>1$ at every input and on the correct training signs. As $t\downarrow 0$, the posterior probability of the common minimizing sign tends to one at every input. Consequently,*

$$\lim_{t\downarrow 0}\mathbb{P}_x\left\{\mathbb{P}\{f(x)Z_t(x)>0\mid \textit{conditioning}\}\le 2/3\right\}\le\delta.$$

The statement concerns sign probabilities, not the Gaussian posterior mean. The variational problem (1) is also the one that arises in the width-first subcritical neural regime $0<\gamma<1$. The proofs below are entirely finite dimensional and use its explicit kernel; they do not establish or require a neural width-limit theorem.

2 Fixed-degree eigenvalues

For $S\subseteq[d]$, write $\chi_S(x)=\prod_{i\in S}x_i$. Let $\lambda_j(d)$ denote the eigenvalue on degree-j characters of the *probability-normalized* kernel operator $2^{-d}K_d$. Thus

$$\|z\|_K^2=\sum_{S\subseteq[d]}\frac{\widehat{z}(S)^2}{\lambda_{|S|}(d)}. \tag{2}$$

Lemma 3. *The matrix K_d is strictly positive definite. For every fixed positive odd integer j,*

$$\lambda_j(d)=F^{(j)}(0)(d+1)^{-j}+O_j(d^{-j-1}),\qquad F^{(j)}(0)>0. \tag{3}$$

The implicit constant may depend on L,j. Moreover, for a constant C_L and $B_k=(C_Lk)^{C_Lk}$, for every $d\ge k$ and odd $1\le j\le k$,

$$B_k^{-1}d^{-j}\le\lambda_j(d)\le B_kd^{-j}. \tag{4}$$

Proof. The odd Taylor series of Ψ has strictly positive coefficients at every odd degree. Composition preserves this property. Expand $F(t)=\sum_{m\text{ odd}}c_mt^m$. The polynomial kernel $((1+x\cdot y)/(d+1))^m$ has nonnegative Walsh eigenvalues. For any character of reduced degree j, the term $m=j$ if j is odd, and the term $m=j+1$ if j is even, gives a strictly positive coefficient: in the latter case use the additional constant coordinate once. This proves strict positive definiteness.

For odd j, write

$$\lambda_j(d)=\mathbb{E}\left[F\left(\frac{1+\sum_{i=1}^dX_i}{d+1}\right)\prod_{i=1}^jX_i\right]$$

with independent uniform signs X_i. Taylor-expand F through degree $2j+1$ on $[-1/2,1/2]$. The complementary event has exponentially small probability; both F and the fixed Taylor polynomial are bounded on $[-1,1]$. Oddness makes the remainder $O_j(|t|^{2j+3})$, so it contributes $O_j(d^{-j-3/2})$, by the fixed moments of a Rademacher sum. Terms of degree less than j vanish. The degree-j term contributes $c_jj!/(d+1)^j$. For a fixed $m>j$, the signed moment of $(1+\sum X_i)^m$ is $O_j(d^{(m-j)/2})$: each of the specified j coordinates must occur an odd number of times, while

other coordinates must occur an even number of times. Since the next odd m is $j+2$, all remaining Taylor terms are $O_j(d^{-j-1})$.

For the uniform bounds, every Taylor coefficient of F is at most one, because they are nonnegative and sum to $F(1)=1$. At odd degrees, $c_j(F) \ge (2/\pi)^{L-2} c_j(\Psi) \ge c_L j^{-3/2}$, by keeping the linear term in each outer composition. The degree-j term alone proves the lower bound in (4), absorbing 2^j and the fixed constants into B_k.

For the upper bound, Taylor-expand through degree $2k+1$. For each odd $m \le 2k+1$ with $m \ge j$, the preceding signed-moment count is bounded by $(Ck)^{Ck} d^{(m-j)/2}$: at most $(m-j)/2$ distinct unspecified coordinates can occur, and all multiplicities and orderings cost at most $(Ck)^{Ck}$. Summing the finitely many terms gives $B_k d^{-j}$. On $|t| \le 1/2$, oddness and the coefficient bound give remainder at most $2|t|^{2k+3}$; its expectation is at most $(Ck)^{k+3/2} d^{-k-3/2}$. On the complementary event the error is at most Ck, with probability at most Ce^{-cd}. Finally, $\sup_{d>0} d^k e^{-cd} \le (Ck)^k$. These errors are absorbed into $B_k d^{-j}$, after increasing C_L. □

3 A globally separated convex minimizer

Choose r to be the smallest integer satisfying

$$r \ge \sqrt{k/\delta}, \qquad r \equiv k \pmod 4. \tag{5}$$

For sufficiently large odd k, r is odd and $r<k$. Set $h_0=(k-r)/2$, which is even. If $H(x)$ is the number of negative active coordinates, let

$$C = \{x : h_0 \le H(x) \le h_0 + r\},$$

including every nuisance-coordinate assignment. Chebyshev's inequality, using $\mathrm{Var}(H)=k/4$, gives

$$\mathbb{P}(C^c) = \mathbb{P}\{|H - k/2| > r/2\} \le k/r^2 \le \delta. \tag{6}$$

Initially impose only the convex constraints on C:

$$z_d = \arg\min_z \{\|z\|_K^2 : f(x)z(x) \ge 1 \text{ for } x \in C\}. \tag{7}$$

Let p_r be the degree-r polynomial interpolating $p_r(i) = (-1)^i$, $0 \le i \le r$, and define

$$P(x) = (-1)^{h_0} p_r(H(x) - h_0). \tag{8}$$

It has degree r on the active cube and training margins exactly one.

Lemma 4. *There is a constant C_L such that, for $B_k = (C_L k)^{C_L k}$ and all $d \ge k$,*

$$\max_x |z_d(x) - P(x)| \le B_k/d, \qquad E_d := \|z_d\|_K^2 \le B_k d^r.$$

Consequently, for $d \ge 2B_k$, z_d is globally separated, is the unique minimizer of (1) with $T=C$, and has error at most δ.

Proof. Strict convexity gives uniqueness in (7). Invariance under nuisance flips makes z_d independent of nuisance coordinates; active permutations make it a function of H. Since k is odd and the layer interval is symmetric, the operation $z(x) \mapsto -z(-x_{[k]}, x_{[d]\setminus[k]})$ also preserves the problem. Thus z_d is odd under $H \mapsto k-H$ and has only odd active Walsh levels. The same symmetry holds for P, by uniqueness of interpolation.

In this proof, B denotes a quantity bounded by $(C_L k)^{C_L k}$; the constant C_L can increase a fixed number of times. Products, fixed powers, and factors such as $k!$ or 2^k are absorbed into this convention. Use the fixed orthonormal symmetric active Walsh basis

$$\phi_j = \binom{k}{j}^{-1/2} \sum_{|S|=j,\ S\subseteq[k]} \chi_S, \qquad j \text{ odd}.$$

These are polynomials of degree j in H. Their leading coefficients and $\max_H |P(H)|$ are bounded by B: for the latter, use $p_r(t) = \sum_{j=0}^{r} (-2)^j \binom{t}{j}$ and $|t| \le k$. Write $c_j(z)$ for the coefficient of ϕ_j. By (4), the comparator P gives

$$\|z_d\|_K^2 \le \|P\|_K^2 \le B d^r.$$

Let u_d be the degree-at-most-r part of z_d, and $v_d = z_d - u_d$. Since the first possible higher degree is $r+2$, the same spectral bounds and $\|v_d\|_\infty \le 2^{k/2}\|v_d\|_2$ give

$$\|v_d\|_\infty \le B/d. \tag{9}$$

A direct coefficient comparison is stronger than merely bounding the energy of u_d. Namely,

$$\begin{aligned} |c_r(z_d)|^2 &\le \lambda_r(d)\|P\|_K^2 \\ &= |c_r(P)|^2 + \sum_{j<r,\ j \text{ odd}} \frac{\lambda_r(d)}{\lambda_j(d)} |c_j(P)|^2 \le |c_r(P)|^2 + B/d^2. \end{aligned}$$

Consequently

$$|\operatorname{lead}(u_d)| \le |\operatorname{lead}(P)| + B/d. \tag{10}$$

Put $y_{i,d} = (-1)^{h_0+i} z_d(h_0+i) \ge 1$. Interpolation at consecutive nodes and (9) give

$$\operatorname{lead}(u_d) = (-1)^{h_0+r} \sum_{i=0}^{r} \frac{y_{i,d}}{i!(r-i)!} + e_d, \qquad |e_d| \le B/d.$$

All these weights are positive, and $|\operatorname{lead}(P)| = \sum_i 1/[i!(r-i)!]$. Thus (10) implies

$$\sum_{i=0}^{r} \frac{y_{i,d} - 1}{i!(r-i)!} \le B/d.$$

Each summand is nonnegative and each weight is at least $1/r!$, so $0 \le y_{i,d} - 1 \le B/d$ for every i. The Lagrange interpolation operator from these $r+1$ consecutive nodes to all $H \in \{0, \dots, k\}$ has norm at most $(r+1)k^r \le B$: each denominator is a nonzero integer and each numerator has r factors of magnitude at most k. Combining this with (9) proves $\|z_d - P\|_\infty \le B/d$.

There is a root of p_r in each interval $(i, i+1)$, $0 \le i < r$; these exhaust its roots. Its absolute value increases as one moves outward from $[0, r]$. Newton interpolation gives $p_r(r+1) = 1 - 2^{r+1}$ for odd r, and reflection gives the same magnitude at $t = -1$. Consequently $|P| \ge 2^{r+1} - 1$ on every integer layer outside C. If $d \ge 2B_k$, the uniform difference is at most $1/2$, so $|z_d| > 1$ outside C; on C, its prescribed margins are already at least one. Thus z_d is feasible for the global problem. Since (7) is a relaxation, it is the unique global minimizer as well. Equation (6) bounds its error. □

Lemma 5 (A large correct amplitude)**.** *At the active input with $H = 0$, the target and P are positive and*

$$P(0) \ge \left(\frac{k-r}{r}\right)^r.$$

Under the dimension condition $d \ge 2B_k$ from Lemma 4, $z_d(0) \ge P(0)/2$.

Proof. The leading coefficient of p_r is $(-2)^r/r!$. All roots of P, viewed as a polynomial in H, lie between h_0 and $h_0 + r$. Since h_0 is even and r is odd, $P(0) > 0$. Therefore

$$P(0) \ge \frac{2^r h_0^r}{r!} \ge \left(\frac{2h_0}{r}\right)^r.$$

The final claim follows from Lemma 4. □

4 Sparsifying the selected examples

The next lemma applies to any finite positive definite covariance. It controls all global minimizers, even when the globally separated feasible set is nonconvex.

Lemma 6 (Sparse teaching by contact sampling)**.** *Let $K(x,x) \le 1$. Suppose z_* is the convex minimum-energy solution of unit-margin constraints on a set C and also satisfies $|z_*| \ge 1$ everywhere. Write $E = \|z_*\|_K^2$. For every positive integer n, there is a multiset T of n contact points in C such that every minimum-energy globally separated field z_T consistent with their prescribed labels satisfies*

$$\|z_T - z_*\|_K \le \frac{2E}{\sqrt{n}}, \qquad \max_x |z_T(x) - z_*(x)| \le \frac{2E}{\sqrt{n}}. \tag{11}$$

Repeated points may be discarded.

Proof. Convex duality and complementary slackness give multipliers $\mu_x \ge 0$ supported on contacts, with

$$z_* = Kh, \qquad h_x = f(x)\mu_x, \qquad \sum_x \mu_x = E, \qquad h^\mathsf{T} K h = E.$$

Draw $X_1, \dots, X_n$ independently with law μ/E, and set

$$h_T = \frac{E}{n} \sum_{i=1}^{n} f(X_i) e_{X_i}.$$

It is a feasible dual vector for the convex margin problem on T, with objective $D_T = 2E - h_T^\mathsf{T} K h_T$. Independence and the diagonal bound imply

$$\mathbb{E}[h_T^\mathsf{T} K h_T] \le (1 - 1/n)E + E^2/n.$$

Some realization therefore has $D_T \ge E - E^2/n$. Let v_T be the convex minimum-norm field satisfying just the training margins on this realization, and let $E_T = \|v_T\|_K^2$. Weak duality gives $E_T \ge E - E^2/n$.

Both z_* and every globally separated minimizer z_T are feasible for this convex problem. Also $\|z_T\|_K^2 \le E$, because z_* is a feasible globally separated comparator. The Hilbert projection inequality at v_T consequently gives

$$\|z_* - v_T\|_K^2 \le E - E_T, \qquad \|z_T - v_T\|_K^2 \le E - E_T.$$

The triangle inequality proves the first claim in (11). The reproducing property and $K(x,x) \le 1$ prove the second. □

Proof of Theorem 1. Increase C_L in the theorem so that Lemma 4 applies for every permitted d. Use Lemma 6 with $z_* = z_d$ and $n = \lceil 16 E_d^2 \rceil$. Every global minimizer on the resulting selected set is within $1/2$ of z_d at every input. Since $|z_d| \ge 1$, all have its sign function, hence error at most δ. At $H = 0$ their amplitude is at least $P(0)/2 - 1/2$, which is $\exp\{c_\delta \sqrt{k} \log k\}$ for sufficiently large k by Lemma 5 and (5).

The energy bound gives $n \le 17 B_k^2 d^{2r}$. Increasing C_L once more ensures that the dimension condition absorbs $17 B_k^2$ into one power of d. Thus $|T| \le n \le d^{2r+1}$. Since $r < \sqrt{k/\delta} + 4$, this is bounded by the power stated in the theorem. □

Proof of Corollary 2. There are finitely many sign orthants. The closed feasible set in each compatible orthant has a minimum Gaussian rate $\|z\|_K^2/2$, by coercivity. All global minimizers have the common sign from Theorem 1, so every other compatible orthant has a strictly larger

minimum rate. Taking the minimum of finitely many positive gaps still gives a positive gap. Strict inequalities in the conditioning do not change the rate infimum: scaling a closed feasible field by $1 + \varepsilon$ makes every absolute margin strict. Elementary finite-dimensional Gaussian Laplace bounds therefore make the total posterior probability of the other orthants tend to zero. The cube is finite, so this gives the stated pointwise probability and confidence conclusion. □

5 Why this does not answer the random-sample problem

The contact sampling in Lemma 6 selects only inputs where the dense convex minimizer already agrees with parity. Uniform sampling also produces examples in its error region. Then that field is no longer feasible, and the upper energy bound $\|z_T\|_K^2 \le E$ used in the sparsification proof fails. Reweighting uniform observations can approximate the contact dual, but it cannot discard those conflicting constraints.

Accordingly, the theorem rules out a $d^{\Omega(k)}$ requirement that is claimed for *every* parity-consistent training set. It neither proves nor disproves the natural lower bound for independent uniform examples. It also rules out a correct-amplitude ceiling derived solely from global optimality and the contact equations. Any such control in the uniform sampling problem must use its probabilistic structure.